

REPUBLIC OF THE PHILIPPINES · DEPARTMENT OF EDUCATION

# Grade 6 Mathematics

## THIRD TERM EXAMINATION

SY 2026–2027 · MATATAG

50 points · Plane Figures, Circles, Composite Figures, Data & Probability  
Includes Table of Specifications and Answer Key for the teacher

# Third Term Examination

Name: \_\_\_\_\_ Section: \_\_\_\_\_ Date: \_\_\_\_\_ Score: \_\_\_\_\_/50

Use  $\pi = 3.14$  where needed. Show solutions for the problem-solving part.

---

## I. Multiple Choice (1 pt each, 20 pts)

---

1. A triangle with no equal sides is  
A. equilateral      B. isosceles      C. scalene      D. right
2. The sum of the angles of a triangle is  
A.  $90^\circ$       B.  $180^\circ$       C.  $270^\circ$       D.  $360^\circ$
3. A quadrilateral with 4 equal sides and 4 right angles is a  
A. rectangle      B. rhombus      C. square      D. trapezoid
4. The sum of the angles of a quadrilateral is  
A.  $180^\circ$       B.  $270^\circ$       C.  $360^\circ$       D.  $540^\circ$
5. Area of a triangle with base 10 and height 6 is  
A. 60      B. 30      C. 16      D. 32
6. Area of a parallelogram, base 8, height 5, is  
A. 13      B. 20      C. 40      D. 80
7. Area of a trapezoid, bases 6 & 10, height 4, is  
A. 16      B. 32      C. 40      D. 64
8. A segment from the center to any point on the circle is the  
A. diameter      B. radius      C. chord      D. arc
9. If  $r = 8$  cm, the diameter is  
A. 4 cm      B. 8 cm      C. 16 cm      D. 64 cm
10. The value of  $\pi$  is approximately  
A. 3.41      B. 3.14      C. 1.34      D. 3.00
11. Circumference of a circle with  $d = 10$  cm is  
A. 31.4 cm      B. 15.7 cm      C. 62.8 cm      D. 78.5 cm
12. The formula for the area of a circle is  
A.  $\pi d$       B.  $2\pi r$       C.  $\pi r^2$       D.  $\pi r$
13. Area of a circle with  $r = 5$  cm is  
A. 31.4      B. 78.5      C. 15.7      D. 25
14. The most frequent value in a data set is the  
A. mean      B. median      C. mode      D. range
15. The middle value of an ordered data set is the  
A. mean      B. median      C. mode      D. total

16. The mean of 4, 6, 8, 10 is

A. 6

B. 7

C. 8

D. 28

17. A pie graph best shows

A. change over time

B. parts of a whole

C. exact values

D. growth

18. P(rolling a 2 on a die) is

A.  $\frac{1}{2}$

B.  $\frac{1}{6}$

C.  $\frac{2}{6}$

D.  $\frac{1}{3}$

19. An event that is sure to happen has probability

A. 0

B.  $\frac{1}{2}$

C. 1

D. 100

20. P(heads on a fair coin) is

A. 0

B.  $\frac{1}{2}$

C. 1

D.  $\frac{1}{4}$

## II. Computation (2 pts each, 16 pts)

---

21. Find the third angle of a triangle with angles  $40^\circ$  and  $75^\circ$ . = \_\_\_\_\_

22. Area of a trapezoid: bases 9 & 15 cm, height 6 cm. = \_\_\_\_\_

23. Circumference of a circle with  $r = 9$  cm. = \_\_\_\_\_

24. Area of a circle with  $d = 12$  cm. = \_\_\_\_\_

25. Mean, median, and mode of 4, 8, 8, 10, 10, 10. = \_\_\_\_\_

26. Angle for 15 of 60 pupils in a pie graph. = \_\_\_\_\_

27. P(number > 4 on a die). = \_\_\_\_\_

28. Area of a parallelogram, base 14 cm, height 9 cm. = \_\_\_\_\_

## III. Problem Solving (show solutions, 14 pts)

---

29. (3 pts) A triangular sail has a base of 4 m and a height of 6 m. What is its area?

30. (3 pts) A circular fishpond has a radius of 10 m. Find its circumference and its area.

31. (4 pts) A figure is a  $12\text{ cm} \times 8\text{ cm}$  rectangle with a half-circle of radius 4 cm attached to one short side. Find the total area.

32. (4 pts) The scores of 5 pupils are 12, 15, 15, 18, 20. Find the mean, median, and mode. A marble is then drawn from a bag of 3 red and 2 white — what is  $P(\text{red})$ ?

# Table of Specifications & Answer Key

Third Term Examination · 50 points

## Table of Specifications

| Competency (code)                                   | Items                       | Pts       | %           |
|-----------------------------------------------------|-----------------------------|-----------|-------------|
| Properties of triangles/quadrilaterals (6MG-IIIa-1) | 1–4, 21                     | 6         | 12%         |
| Perimeter & area of plane figures (6MG-IIIb-2)      | 5, 6, 7, 22, 28, 29         | 11        | 22%         |
| Circles: parts & circumference (6MG-IIIc-3)         | 8, 9, 10, 11, 23, 30        | 9         | 18%         |
| Area of circle & composite (6MG-IIId-4)             | 12, 13, 24, 31              | 8         | 16%         |
| Data & central tendency (6DP-IIIf-5, IIIf-6)        | 14, 15, 16, 17, 25, 26, 32a | 10        | 20%         |
| Simple probability (6DP-IIIf-7)                     | 18, 19, 20, 27, 32b         | 6         | 12%         |
| <b>Total</b>                                        | <b>32 items</b>             | <b>50</b> | <b>100%</b> |

## Answer Key — Part I (Multiple Choice)

| 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
|----|----|----|----|----|----|----|----|----|----|
| C  | B  | C  | C  | B  | C  | B  | B  | C  | B  |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| A  | C  | B  | C  | B  | B  | B  | B  | C  | B  |

#7:  $\frac{1}{2}(6+10) \times 4 = 32$  (B). #13:  $3.14 \times 25 = 78.5$  (B).

## Answer Key — Part II (Computation)

21)  $65^\circ$  22)  $\frac{1}{2}(9+15) \times 6 = 72 \text{ cm}^2$  23)  $2 \times 3.14 \times 9 = 56.52 \text{ cm}$  24)  $r=6 \rightarrow 3.14 \times 36 = 113.04 \text{ cm}^2$  25) mean = 8.33, median = 9, mode = 10 26)  $15 \div 60 \times 360 = 90^\circ$  27)  $\frac{2}{6} = \frac{1}{3}$  28)  $126 \text{ cm}^2$

## Answer Key — Part III (Problem Solving)

29)  $A = \frac{1}{2} \times 4 \times 6 = 12 \text{ m}^2$ .  
 30)  $C = 2 \times 3.14 \times 10 = 62.8 \text{ m}$ ;  $A = 3.14 \times 10 \times 10 = 314 \text{ m}^2$ .  
 31) rectangle  $12 \times 8 = 96$ ; half-circle  $= \frac{1}{2} \times 3.14 \times 4^2 = 25.12$ ; total = **121.12 cm<sup>2</sup>**.  
 32) mean = 16, median = 15, mode = 15;  $P(\text{red}) = \frac{3}{5}$ .